

FLEA NEWS 36

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Please find herewith:

- Fleas in Literature: a Polish poem and a continuation of the quotations from early authors
- Book reviews
- Lists of additional literature for 1981 (list 14), 1982 (list 11), 1984 (list 9), 1985 (list 7), 1986 (list 5), 1987 (list 3) and 1988 (list 1)
- Additions, changes and deletions from the mailing list

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RECENTLY PUBLISHED

An illustrated catalogue of the Rothschild collection of fleas (Siphonaptera) in the British Museum (Natural History). Volume VII. Malacopsylloidea (Malacopsyllidae and Rhopalopsyllidae) by F. G. A. M. Smit, 1987. ISBN 0-19-858519-5. Oxford University Press/The British Museum (Natural History), Oxford & London. 380 pp. + 5 pls., 710 figs. (including 119 maps), £75.00, US\$152.11. (According to the author, the exact day of release was 12.XI.1987.)

Over the years students of the Siphonaptera have grown accustomed to the high quality of the volumes in the Catalogue series produced by the British Museum (Natural History). Volume VII is perhaps the masterpiece of the entire series for a number of reasons. Although it follows the same general format as the preceding volumes, it differs in a significant number of ways. While earlier volumes contained lengthy prose in the sections dealing with the various taxa, illustrations from a number of sources, and no maps, all of these irregularities have been eliminated from this volume.

Brief introduction and acknowledgment sections are followed by a "List of Genera and Subgenera of Siphonaptera" arranged alphabetically by genus, followed by the family and subfamily where appropriate, arranged according to the system of classification first presented by Smit in 1982 and discussed in Flea News 31: 183-186 (December 1985). The introductory section is concluded with a list of taxonomic changes and new taxa described in the volume, and abbreviations of type depositories and the names of collectors.

The bulk of the volume consists of taxonomic treatments of the taxa from superfamily to subspecies. Unlike preceding volumes, this tome contains an

extensive bibliography. As a result, the synonymic sections of the supraspecific taxa are very much abbreviated, containing only the names of authors, date of publication and an initial page number. The species treatments follow much the same form as in the earlier volumes in some instances, in abbreviated form in others. Discussion of the anatomical characters is usually limited to references to the figure numbers of these structures, followed by a distributional listing (including a map number) and a list of hosts. In some cases there may be a brief commentary about points of importance concerning the species, and each section is concluded with a list of material in the collection if it is represented in the British Museum (Natural History).

The volume concludes with a gazetteer of collection localities, including map coordinates where available, and the species collected there, a host-flea index, an extensive bibliography, a genus and species index and five photographic plates containing 38 figures.

This volume is a fitting tribute to its author and his more than 40 years of devotion to the study of the order Siphonaptera. Mr. Smit is to be congratulated for his pursuit of excellence as embodied in this publication. ---R.E.L.

Male Genitalia and the Phylogeny of the Pulicoidea (Siphonaptera) by Thomas B. Cheetham, 1988. Theses Zoologicae, Volume 8. Ronald Fricke, editor. Koeltz Scientific Books, D-6240 Koenigstein, W. Germany. i-vii, 224 pp., 1 text figure + 86 pls. (Price not available at present.)

This is the publication, in toto, of the doctoral dissertation of the author, albeit containing a few minor alterations and a slightly different arrangement. After a brief introductory section including materials and methods and acknowledgments, the text is divided into two parts. Part I, entitled "Development and anatomy of the male genitalia in Xenopsylla cheopis (Rothschild, 1903)" deals with the basic genitalic anatomy and development from the early larval to the adult stage. Adult male genitalia are discussed in some detail and the section ends with a summary of abdominal segmentation and the origin of male genitalia in the order, including the origin of the claspers and primary phallic lobes. The major segmental limits of segments eight through eleven are also discussed, thus setting the stage for the treatment in Part II.

Part II first deals with male genitalic anatomy for all of the pulicoid genera where this sex is known. The second section of Part II contains subsections dealing with : a) Relationships among pulicid fleas, including character analysis, a discussion of plesiomorphic characters and the erection of major subdivisions to include the seven subfamilies. This is followed by b) a discussion of cladistic relationships in the Pulicoidea and a cladogram of the taxa, c) a consideration of phylogeny and zoogeography within the group, and d) a discussion of the homologies of some male structures.

The book is terminated by a general summary and bibliography and two appendices. Appendix I is a species list for the Pulicidae, including host association and distribution. Appendix II is a species list for the genus Xenopsylla arranged according to the traditional species groups set down in

Hopkins & Rothschild (1953). The remainder of the book is comprised of 86 plates consisting of line drawings and light and scanning microscope photographs of the pulcid genera showing the relevant details of the male genitalia.

This is a remarkable piece of work, and the first of its kind for any group of taxa in the order Siphonaptera. Through the cooperation of many colleagues throughout the world it was possible to obtain material of every genus where males are known, the exception being Nesolagobius, known only from females. The success of the study is itself a tribute to their cooperation, and the author is to be commended for his skill and tenacity in performing this analysis.

I am informed that a limited number of copies are available for distribution. Interested parties should contact the author at the following address after 1 September 1988.

Dr. Thomas B. Cheetham
Department of Physical & Life Sciences
Wilson College
Chambersburg, Pennsylvania 17201 USA

---R.E.L.

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INFLATION - In Flea News number 16 (IX-1979) F. G. A. M. Smit lamented the recent price increase of the Hopkins & Rothschild volumes. Following are quotes from the latest publications catalogue from the British Museum (Natural History) for the 7 volumes, accompanied by the 1979 prices in parentheses. Prices are in Sterling pounds. Volume I, 60.00 (13.00); II, 60.00 (16.50); III, 60.00 (19.50); IV, 60.00 (28.50); V, 60.00 (34.00); VI, 60.00; VII, 75.00. Totals: 1979, 111.50 for 5 volumes; 1988, 435.00 for 7 volumes. That is an average increase of 390% in less than 10 years (269% on the first 5 volumes). Lamentations are indeed in order.

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CONGRATULATIONS!

DR. EVERETT WILLIAMS JAMESON, JR. is retiring from the Zoology faculty of the University of California, Davis, after 40 years of service in that department. A native of Buffalo, New York, "Bill" received B.S. and Ph.D. degrees from Cornell University, Ithaca, New York, and an M.A. degree from the University of Kansas, Lawrence, before assuming the position at Davis. Although trained as a vertebrate zoologist, many of his publications have dealt with the fleas and ectoparasitic mites of small mammals. In 1962 he coauthored "The Siphonaptera of Japan" with the late Kôhei Sakaguti, as Pacific Insects Monograph #3.

Bill's Siphonaptera collection consists of approximately 5000 microscope mounts and is particularly rich in material from Taiwan, Japan, Korea and western United States. It has been incorporated into the Lewis Collection here at Iowa State University and will ultimately become part of the Siphonaptera collection at the Field Museum of Natural History in Chicago, Illinois.

When asked about his retirement plans, he indicated that he had a number of projects dealing with western mammals and their biology which he intended to pursue.

The editors of Flea News join his colleagues in wishing him a long and productive retirement. His mailing address is now 13 Oakside, Davis, California 95616 USA. ---R.E.L.

DR. CLUFF E. HOPLA, George Lynn Cross Research Professor of Zoology at the University of Oklahoma, Norman, will formally retire on 1-July-1988 after being given emeritus status in January of this year. Cluff has been a member of the Department of Zoology at Oklahoma since 1951 and served as department chairman from 1962 to 1970. To pulicologists he is best known for his work with the Arctic Aeromedical Laboratory in Fairbanks, Alaska, which culminated in a monograph on the Siphonaptera of Alaska in 1965. In a recent telephone conversation, Cluff indicated that he would continue to maintain his office and laboratory and extend his studies on the parasites of cliff swallows.

The editors of Flea News would like to take this opportunity to congratulate Cluff Hopla for his years of service to the community and wish him a long and fruitful retirement.

FRANK CLARK, Department of Zoology, The University, Leicester, U.K., has successfully defended his Ph.D. dissertation entitled "Studies on three congeneric species of flea from the nests of Delichon urbica urbica in England". ---J.H.L.

MICHAEL DRYDEN, DVM, School of Veterinary Medicine, Purdue University, West Lafayette, Indiana, has completed work on his Masters degree. His thesis, which is on file at the Purdue University Library, is entitled "Evaluation of certain parameters in the bionomics of Ctenocephalides felis felis (Bouché, 1835)". He will continue his cat flea bionomics studies while working for his Ph.D. at Purdue. ---J.H.L.

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CONFERENCES

At the 40th annual Midwestern Conference of Parasitologists held June 2-4, 1988, at Purdue University, West Lafayette, Indiana, a paper on fleas was presented:

"Winter acclimation, survival and supercooling point depression in the bird flea, Ceratophyllus idius (Siphonaptera)."

Douglas Schelhaas, Biology Department, U.S. Air Force Academy, Colorado Springs, Colorado, and

Omer Larson, Biology Department, University of North Dakota, Grand Forks.

A symposium on the topic "Results and Prospects of Research on the Siphonaptera in the Palaearctic from the aspect of their Significance in Practice" was held June 6-11, 1988, in Bratislava, Czechoslovakia. The symposium, sponsored by the Slovak Entomological Society of the Slovak Academy of Sciences and the Department of Zoology, Comenius University, Bratislava, was headed by Academician Bohumír Rosický with the assistance of Doc.RNDR. D. Matis and RNDR. M. Kiefer. ---J.H.L.

PLAGUE IN THE UNITED STATES. During a recent conversation with Dr. Allan Barnes, Chief of the Plague Branch of the Public Health Service, Centers for Disease Control, Fort Collins, Colorado, he reported that only three cases of plague have been diagnosed so far in 1988. The first occurred in February in Pecos County, western Texas, and the remaining two were diagnosed in June from Costilla County, Colorado, and McKinley County, New Mexico. All of the victims made satisfactory recoveries. Dr. Bernard Nelson of the California Department of Health Services, reporting in the latest Vector Ecology Newsletter, states that plague has been detected in wild rodents, wild carnivores and domestic pets in five counties in California, but no human cases have been diagnosed during 1988. ---R.E.L.

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NOTES

TO DATE there have only been three responses to our offer of the species lists mentioned on pages 256-258 of Flea News 35. This is insufficient to justify the duplication process and we must await a larger response before we proceed. ---R.E.L.

SOME RECIPIENTS OF FLEA NEWS 36 will note that we have included reprints of Smit, 1981, "The song of the flea....". Copies have been sent only to those whose names are on our regular reprint mailing list. If you are interested in this paper but did not receive it, a limited number of copies are still available. ---R.E.L.

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FLEAS IN LITERATURE

Following is a continuation of quotations from some early authors prepared by F. G. A. M. Smit. Readers are referred to Flea News 35, p. 259-261, for the first installment which includes all the references to the quotations.

HEAD

They have a little head. (Moufet 1634: 275; 1658: 1101. Jonston 1653)
He [the flea] has also two pointers before which grow out of the forehead, by which he tries and feels all objects, whether they be edible or no. (Power 1664: 2)

EYE

It is very probable that they have eyes, both because they choose their places of retreat, and because they withdraw themselves when the day breaks. (Moufet 1634: 276; 1658: 1102)
... with a small head, but in it two fair eyes globular and prominent of the circumference of a spangle; in the midst of which you might (through the diaphanous Cornea) see a round blackish spot, which is the pupil or apple of they eye, beset round with a greenish glistering circle, which is the Iris, (as vibrissant and glorious as a Cats eye) most admirable to behold. How critical is Nature in all her Works! that to so small and contemptible an Animal hath given such an exquisite fabrick of the eye, even to the distinction of parts. Had our famous Muffet [Moufet] but seen them, he could not have spoke so doubtfully as he did: Oculos (saith he, speaking of Flea's) habere, verisimile est, tùm quòd suos eligunt recessus, tùm quòd appetente luce se subjuçant.

...that he might not be hurt by the great leaps he takes; to which purpose also he hath so excellent an eye, the better to look before he leap; ... (Power 1664: 1-2)

MOUTHPARTS

The point of his nib is something hard, that he may make it enter the better. It must necessarily be hollow, that he may suck out the blood, and carry it in. (Moufet 1634: 276; 1658: 1101-1102)

At his snout is fixed a Proboscis, or hollow trunk or probe, by which he both punches the skin, and sucks the blood through it, leaving that central spot in the middle of the Flea-biting, where the probe entered. (Power 1664: 2)

THORAX

... with a very short neck. (Moufet 1634: 275; 1658: 1101)

He [the flea] has also a very long neck, jemmar'd like the tail of a Lobster, which he could nimbly move any way: ... (Power 1664: 2)

LEGS

Their hinder little legs are bent backwards toward their bellies, and their forelegs toward their breasts, as four-footed beasts are, as it is usual almost in all Insects to whom Nature hath given but four feet. It may be for that end the joynts of Fleas are so disposed, that they may with the more ease hide themselves in the long foldings and plights of the blankets from those that hunt after them. The ends of their feet are divided into two parts, and are hooked and sharp, and seem as it were to be horny, not only that they may more surely creep up upon high places, but also that they may sit and stick faster to the smooth skin. (Moufet 1634: 275; 1658: 1101)

... advantageous contrivance of the joynts of his hinder legs which bend backwards towards his belly, and the knees or flexure of his fore-legs forwards (as in most quadrupeds) that he thereby take a better rise when he leaps. His feet are slit into claws or talons, that he might the better stick to what he lights upon; ... (Power 1664: 2)

FEEDING

Fleas suck so much blood that they pass it out continually blackish and dry. (Magnus 1255-1270)

Fleas feed on dust; they are first white, then black and desire blood. With their sharp biting mouthparts they even attack kings. Especially painful is their bite before rain. (B. Anglicus 1483)

They seek for the most tender places, and will not attempt the harder places with their nibble; with two very small foreyards that spring out of their foreheads, they both prove their way, and judge of the nature of the object, and whether it be hard or soft: ... The lesser, the leaner, and the younger they are, the sharper they bite, the fat ones play and tickle men more willingly. In the morning, after they have fed, they creep into the rough blankets, and stick to the walls, or else they hide themselves in the rushes or dust; ... (Moufet 1634: 276; 1658: 1102)

EFFECT OF BITES

Where they bite they leave a red spot as a Trophie of their force, which they set up. (Moufet 1634: 276; 1658: 1102)

If one is not able to keep watch or sleep, it [the flea] will produce a numbness in the limbs. (Cantimpré 1233-1248; Anonymous 1491)

ALIMENTARY CANAL

They have but one small intestine with folds inward, which is either relaxed or contracted as they eat more or lesse. (Moufet 1634: 276; 1658: 1102)

Professor W. Skuratowicz kindly sent the following poem which we would like to share with our readers. It is from the volume *Modlitwy Zwierząt* (Animal Prayers) by Carmen Bernos de Gasztold, published in Poznań, Poland, in 1987. The translation was provided by Dr. Barbara von Wittich. Emeritus Professor, Department of Foreign Languages and Literatures, Iowa State University.

Modlitwa Pchły

PRAYER OF A FLEA

Skaczę i kłuję,
skaczę i kłuję!
Boże, jak mnie to bawi!
To był genialny pomysł
stworzyć mnie tak małą
i uzbroić mnie w susy i skoki.
Skaczę i kłuję,
skaczę i kłuję.
Co za królewska zabawa!
Przyznaję, że robię to
ze szczyptą złośliwości...
W mej szacie o pchlim kolorze
— barwę tę wymyślono dla mnie —
mam większą władzę
od słonia.
Jak się o tym myśli,
można pęknąć ze śmiechu!
Skaczę i kłuję,
skaczę i kłuję!
Panie,
czy przyjmiesz mnie do swego Raju
nie bojąc się,
że mogę tam wszystkim zaleźć za skórę?
Aż nie śmiem powiedzieć...

I jump and bite,
I jump and bite!
O Lord, do I enjoy this!
It was a brilliant idea
to create me so small
and to arm me with jumps and leaps.
I jump and bite,
I jump and bite.
What royal fun!
I admit that I do this
with a bit of malice ...
In my flea-colored robe
- this color was invented for me -
I have more power
than an elephant.
If one thinks about it,
one can burst with laughter!
I jump and bite
I jump and bite!
O Lord,
will you admit me to your Paradise
without being afraid
that I might creep under everybody's skin?
I almost don't dare to say ...

Amen.

Amen.

* * *

REPRINTS RECEIVED - We wish to thank the following for supplying reprints and citations since the publication of *Flea News* 35:

A. F. Azad	T. Galloway	K. Poole	R. Summerfelt
D. Cyprich	M. Greenwood	J. Segerman	R. Timm
J. Dinsmore	H. Katz	W. Skuratowicz	D. Q. Wang
A. Dudich	J. Keirans	F. Smit	Z. Wegner
L. Durden			

* * *

MAILING LIST

ADDITIONS

Dr. Jalil Abul-hab, Medical Entomology Division, Regional Training Centre for Malaria & Vector Biology & Control, P.O. Box 683, Alwiyah, Baghdad, Iraq

ADDITIONS - contd.

Ruth Sponsler, 1202 Wallace Avenue, Opelika, Alabama 36801 USA

Dr. G. B. White, 105 Breamwater Gardens, Ham, Richmond, Surrey TW10 7SG,
England

CHANGES

Dr. L. A. Durden, Siphonaptera Section, Department of Entomology, NHB 543 MRC
NHB 165, National Museum of Natural History, Washington, D.C. 20560 USA

Dr. Zofia Wegner, Instytut Medycyny Morskiej i Tropikalnej, Zakład Parazytologii
Tropikalnej, 9b Powstania Styczniowego Street, 81-519 Gdynia, Poland

DELETIONS

Dr. N. Haarløv (deceased)

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ADDENDA ET ERRATA

Please note that the Beaucournu citation on page 267 of Flea News 35 should
read Beaucournu, J.-C. & H. Launay - 10 Arthropodes (II)

Also, in the provisional list of taxa on page 263, Ctenophthalmus (C.)
bifidatus mancus was described by Beaucournu & Orhan, not Orhan & Beaucournu,
and Tetrapsyllus satyrus Beaucournu & Torres-Mura should be added to 1986.

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A PREFATORY NOTE TO THE SPECIES LIST - In keeping with Mr. Smit's
practice of numbering the taxa chronologically, the following list includes
numbers for those taxa described in 1986. Readers are urged to call any
omissions in these lists to our attention.

- 98601 allisoni Mardon Coorilla
- 98602 ussuriensis Medvedev Ischnopsyllus (Hexactenopsylla)
- 98603 superjecta Zhang, Liu & Zhang Frontopsylla (Frontopsylla) wagneri
- 98604 concavus Liu, Liu & Zhang Paradoxopsyllus jinshajiangensis
- 98605 affinis Li Lentistivalius
- 98606 diana Beaucournu, Gallardo & Launay Plocopsylla
- 98607 deqinensis Xie & Li Neopsylla affinis
- 98608 mancus Beaucournu & Orhan Ctenophthalmus (Ctenophthalmus) bifidatus
- 98609 tvrtkovici Brelih Ctenophthalmus (Mediostenophthalmus) nifetodes
- 98610 petrovi Brelih Ctenophthalmus (Mediostenophthalmus) nifetodes
- 98611 krystufeki Brelih Ctenophthalmus (Mediostenophthalmus) nifetodes
- 98612 dzukici Brelih Ctenophthalmus (Mediostenophthalmus) nifetodes
- 98613 beishanensis Wu, Ni & Wu Callopsylla (Geminopsylla)
- 98614 yunnanensis Li & Yan Parodontis riggenbachi
- 98615 xijiensis Wu, Chen & Li Nearctopsylla (Neochinopsylla)
- 98616 proboscis Wu, Xiue & Hu Ctenophthalmus (Sinostenophthalmus)
- 98617 terrestus Chen, Ji & Wu Ctenophthalmus (Sinostenophthalmus) taiwanus
- 98618 sichuanoyunnana Xie, Chen & Liu Peromyscopsylla himalaica
- 98619 qinghaiensis Liu, Cai & Wu Ornithophaga anomala
- 98620 sichuanensis Liu, Zhai & Liu Geusibia apromina
- 98621 borealosichuana Liu & Zhai Frontopsylla (Frontopsylla) spadix
- 98622 shennongjiaensis Ji, Chen & Liu Frontopsylla (Frontopsylla) spadix

(1981 - 1984)

- 98623 qinghaiensis Liu, Cai & Pan Frontopsylla (Frontopsylla) nakagawai
98624 borealosisinica Liu, Wu & Chang Frontopsylla (Frontopsylla) nakagawai
98625 tiebuensis Chen, Wei & Liu Frontopsylla (Frontopsylla) exilidigita
98626 stenotus Liu, Cai & Wu Paradoxopsyllus
98627 diversus Liu, Chen & Liu Paradoxopsyllus
98628 hainanensis Liu & Pan Macrostylophora hastata
98629 leizhouensis Li, Huang & Liu Nosopsyllus (Nosopsyllus) wualis
98630 fengi Liu, Xie & Wang Ceratophyllus (Monopsyllus)
98631 satyrus Beaucournu & Torres-Mura Tetrapsyllus

PROVISIONAL LIST OF TAXA DESCRIBED IN 1987 (None so far in 1988)

- Caenopsylla janineae Beaucournu & Gouat
Caenopsylla laptevi ibera Beaucournu & Marquez
Amphipsylla longispina gongheensis Zhang & Ma
Stenischia xiei Li
Plocopsylla viracocha Schramm & Lewis
Plocopsylla nungui Schramm & Lewis
Plocopsylla athena Schramm & Lewis
Plocopsylla kilya Schramm & Lewis
Frontopsylla (Frontopsylla) elatoides intermedia Cai, Wu & Zhang
Geusibia falsotorosa Wu, Ni & Li
Ceratophyllus (Monopsyllus) hamatus Cai & Wu
Ceratophyllus (Monopsyllus) forficatus Cai & Wu
Macrostylophora nandanensis Li, Zeng & Zeng
Ectinorus (Ectinorus) nomisis Smit
Ectinorus (Ichyonus) onychius deplexus Smit
Parapsyllus ambrosius Smit
Tiamastus deflatus Smit
Polygenis (Gephyropsylla) klagesi rangeli Smit
Polygenis (Polygenis) trapidoi mendezi Smit

* * *

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 14)

- Whitehead, P. & C. Keats - The British Museum (Natural History). British Museum (Natural History). 128 pp. [Siph. & plague: 91-94]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1982 (list 11)

- Derbot, S. & C. Mermod - Quelques siphonaptères de mustélidés, dont Rhadinosylla pentacantha (Rothschild, 1897) nouvelle espèce pour la Suisse. Rev. suisse Zool. 89(1): 27-32

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1984 (list 9)

- Takahashi, S. - Survey on accidental introductions of insects entering Japan via aircraft. In: Laird, M., ed. Commerce and the spread of pests and disease vectors. Praeger Publ., New York, p. 65-79

(1985 - 1986)

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1985 (list 7)

- Anderson, J. F. - Epidémiologie de la maladie de Lyme dans la Connecticut (U.S.A.). Bull. Soc. Française Parasitol. 1: 53-60
- Coulanges, P. & V. Ravaoalimalala - Service de la Peste. Arch. Inst. Pasteur Madagascar 53(2): 142-162
- Egri, B. & G. Tersztyánszky - Mass occurrence of *Ixodes ricinus* and of *Monopsyllus sciurorum* [*Ceratophyllus sciurorum*] on a red squirrel (*Sciurus vulgaris*) [in Hungary]. Magyar Allatorvosok Lapja 40(9): 550 [In Hungarian, no English summary]
- Haitlinger, R. - Arthropods occurring on *Sciurus vulgaris* in Poland. Polskie Pismo Ent. 55(2): 429-432
- Haitlinger, R. & A. L. Ruprecht - Arthropods collected from Kujawian bats (Acari and Siphonaptera). Polskie Pismo Ent. 55(3): 615-618
- *Paul, A. - An outline of flea control products and practices, 1985. Vet. Prof. Topics, Small Animal Prof. Topics 10(2): 19 pp.
- Ramisz, A. & M. Pietrak - Bericht über den Einsatz von Deltamethrin gegen Ektoparasiten bei Zootieren. Verhandlungsber. Intl. Symp. Erkrankungen Zootiere 27: 493-496

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1986 (list 5)

- *Adams, J. R. & E. T. Schmidtman - Electron microscopy of cat flea, *Ctenocephalides felis* (Bouché), infected with a rickettsial-like organism. In: Samson, R. A., J. M. Vlak & D. Peters (Eds.), Fundamental and applied aspects of invertebrate pathology. Wageningen, The Netherlands, Foundation of the Fourth International Colloquium of Invertebrate Pathology, p. 459
- Ahmed, B. A. & O. M. E. El-Azazy - Ectoparasites of stray dogs in Zagazig City, Sharkia Province. Assiut vet. Med. J. 16(31): 161-166
- Almeida, C. R. de, A. M. P. de Almeida, D. P. Brasil, J. Dantas Sobrinho & M. A. M. Leal - Estudo do roedor *Akodon arviculoides* Wagner, 1842 (Cricetidae) - importância nos focos pestosos do Brasil. Mem. Inst. Oswaldo Cruz 81(4): 409-416
- Anonymous - Diseases subject to the IHR - plague. Wkly Epidem. Rec. 61(50): 389
- Arnaud, P. H. & H. B. Leech - George Pearson Holland 1911-1985. Pan-Pacific Ent. 62(2): 167
- Azad, A. F. - Acquisition and persistence of murine typhus infection in *Xenopsylla cheopis* after feeding on immune rats. In: D. Borovsky & A. Spielman (Eds.), Host regulated developmental mechanisms in vector arthropods. Univ. of Florida-IFAS, Vero Beach, Florida, pp. 158-162
- Bernard, J. & W. Biesemans - Ectoparasites d'oiseaux de Belgique. Bull. Annales Soc. Roy. Belge Entomol. 122(9-12): 269-271
- Bi, L.-f. - An introduction to a high effect method and equipment for absorbing flea. Kunchong Zhishi 23(3): inside back cover following p. 144. [In Chinese, no English summary]

(1986)

- *Carlotti, D. N. & D. Heripret - La dermatite par allergie aux piqûres de puce chez le chien. Pratique Méd. Chirurg. Animal Compagnie 21 (Suppl. 6), 64 pp.
- Guillon, N., N. A. Araújo Filho & I. A. Sherlock - Ectoparasitos de roedores e marsupiais no ambiente silvestre de Ilha Grande, estado do Rio de Janeiro, Brasil. Mem. Inst. Oswaldo Cruz 81(2): 233-234
- Haitlinger, R. - Arthropods occurring on *Sorex alpinus* Schinz (Mammalia, Insectivora) in Poland. Wiadom. Parazyt. 32(4-6): 459-462
- Kaczmarek, S. - Ectoparasites from the nests of the swallows (*Hirundo rustica* L.) and the house martin (*Delichon urbica* (L.)) collected in the vicinity of Szczecin. Wiadom. Parazyt. 32(4-6): 469-472
- Kohn, M. - Effect of large city conditions on the species composition of fleas parasitizing small mammals. Wiad. Parazyt. 32(4-6): 381-384
- Kruminis-Lozowska, W. - Fleas (Siphonaptera) on *Rattus norvegicus* (Berk.) from the cities of Gdynia and Gdańsk. Wiad. Parazyt. 32(4-6): 385-387
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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

* * *

AS AN AFTERTHOUGHT ----

Our book on the fleas of the Pacific Northwest is at the printers and should be available in late September. It is being published by the Oregon State University Press.

Other projects in various stages of completion include the following:

X (Dr. Beth Schramm's Ph.D. dissertation has been accepted for publication as a volume of *Theses Zoologicae* and is currently being entered in our word processor for printing on a laser printer.

We are assembling a catalogue of invalid names in the Siphonaptera, as well as a catalogue of valid names in the order. Additionally we are working toward a monograph of the Siphonaptera of the Nearctic Region.

In responding to our request for an update of our mailing list, take a few minutes off and let us know about your projects. This is your newsletter and it is available for the dissemination of this type of information!

ANOTHER AFTERTHOUGHT ----

The annual conference of the Society for Vector Ecology will be held at the International Resort Hotel, Palm Springs, California, November 16-18, 1988. For further information, please contact B. Fred Beams, Society for Vector Ecology, P.O. Box 87, Santa Ana, California 92702 USA.